

Algebra 3

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1 Rings

1.1 Monoid rings

Definition 1.1.1. A **ring** is an abelian group $(R, +, 0)$ which is also a monoid $(R, \cdot, 1)$ such that $\cdot$ distributes over $+$. We may also require $0 \neq 1$.

Some classical examples are:

- $\mathbb{Z}$, the ring of integers;
- any field like $\mathbb{F}_{p^n}$, $\mathbb{Q}$, $\mathbb{R}$, or $\mathbb{C}$;
- $M_n(R)$ the ring of $n \times n$ matrices with entries in R , the first noncommutative example here;
- R -valued functions out of any set have a pointwise ring structure;
- the first Weyl algebra is roughly “the ring of polynomial valued differential operators”, defined as

$$A_1 := \mathbb{C}[x, \partial] / (\partial x - x\partial = 1)$$

you should think of this acting of $f \in \mathbb{C}[x]$, generated by the product rule:

$$\partial x f - x \partial f = \frac{d}{dx}(x f) - x \frac{d}{dx} f = f.$$

Definition 1.1.2. For a monoid P and a ring R the **monoid ring** is the set of finite R -linear sums:

$$\sum_{p \in P} r_p p.$$

The addition and multiplication come from the independent inclusions $R, P \subset R[P]$ by $r \mapsto r1_p$ and $p \mapsto 1_R p$ respectively.

Obviously, every group is a monoid, but here are some other examples:

- $\mathbb{N}$ is a semiring, so a monoid under addition;
- given a set X , both $\cup$ and $\cap$ make $\mathcal{P}(X)$ a monoid with $\emptyset, X$ the respective identities;
- the set of endomorphisms of an object in a category is always a monoid;
- let $C \subseteq \mathbb{R}^n$ be a convex cone, i.e. $\mathbb{R}_{\geq 0}C = C$ and $C + C = C$, then $P = C \cap \mathbb{Z}^n$ is a monoid under addition, if $C = \langle (1, 0), (-1, 2) \rangle$ is generated by points in $\mathbb{Z}$ we may not necessarily have P generated as a monoid by these same elements (in this case $(0, 1) \in P$ but cannot be realised as a $\mathbb{Z}$ -linear combination of $(1, 0)$ and $(-1, 2)$).

Most of the canonical examples of rings missed out earlier was because they can be realised as monoid rings:

- $R[x_1, \dots, x_n]$ the ring of polynomials in n variables, is just the monoid ring $R[\mathbb{N}^n]$;
- $R[x_1, x_1^{-1}, \dots, x_n, x_n^{-1}]$ the ring of Laurent polynomials in n variables, is the monoid ring $R[\mathbb{Z}^n]$;
- we will see that a lot of ring homomorphisms are induced by monoid homomorphisms they are rings over, a first example of this is the isomorphism $\mathbb{C}[\mathbb{Z}/n\mathbb{Z}] \cong \mathbb{C}[x]/(x^n - 1)$;
- we can define the quaternions $\mathbb{H}$ as a quotient of $\mathbb{R}[Q_8]$, where Q_8 is the quaternionic group $\{\pm 1, \pm i, \pm j, \pm k\}$, by the ideal $(1 + (-1), i + (-i), j + (-j), k + (-k))$;
- $\mathbb{R}[P]$ from earlier is a subring of $\mathbb{R}[x, y]$ which, as P isn't generated by $\{(1, 0), (-1, 2)\}$, is actually $\mathbb{R}[x, y, y^2/x]$.

Definition 1.1.3. For two rings S, R a **ring homomorphism** is a function $f : R \rightarrow S$ such that f is a homomorphism of the additive groups and multiplicative monoids.

Constructing ring is in general very hard, it is a lot of data.

Definition 1.1.4. If R is a ring and $f : P \rightarrow Q$ is a monoid homomorphism then there is an **induced homomorphism** $f_* : R[P] \rightarrow R[Q]$ given by:

$$\sum_{p \in P} r_p p \mapsto \sum_{p \in P} r_p f(p).$$

This makes a lot of interesting ring homomorphisms, anything more interesting belongs to the land of algebraic geometry, we will not discuss that here.

The **image** and **kernel** of a ring homomorphism are inherited from the additive group homomorphism.

Definition 1.1.5. An additive subgroup $I \leq R$ is a **left ideal** if $RI \subseteq I$. Right ideals and two-sided ideals are defined obviously.

The kernel of a ring homomorphism is certainly a two-sided ideal. Conversely, to any ideal I there is a unique ring structure on R/I that makes $\varphi : R \rightarrow R/I$ a ring homomorphism.

Definition 1.1.6. A **left unit** in R is an element $u \in R$ such that there exists some $v \in R$ with $uv = 1$.

Are left units always right units? We know this to be true in commutative rings and $M_{n \times n}(k)$, but if we consider $GL(\mathbb{R}^{\mathbb{N}})$, with the basis $\{e_1, e_2, \dots\}$, then the linear map which sends each e_i to e_{i+1} is injective and has a left inverse, but is not surjective so has no right inverse. But, if u has both a right inverse **and** a left inverse, then these will always be the same. The group of two-sided units is called $R^\times$.

1.2 Classical algebra and whatnot

You probably already know what an Euclidean domain, principal ideal domain, unique factorisation domain, and a Noetherian ring are.

There won't be a particularly rigorous discussion of the relationship, because I don't care. Look at any set of official notes for something of higher quality if you require it.

Proposition 1.2.1. An Euclidean domain is a principal ideal domain.

Proof. Let R be an Euclidean domain with degree function φ , and let $I \trianglelefteq R$, consider $\varphi(I)$, this must have a smallest element, call it a . Now for any $i \in I$, we can write $i = qa + r$ with $\phi(r) < \phi(a)$, but r must be in I so cannot have degree less than that of a , so $r = 0$. Therefore, $I = (a)$. $\square$

Proposition 1.2.2. A ring is Noetherian iff all ideals are finitely generated.

Proof. Suppose an ideal $I \trianglelefteq R$ does not admit a finite generating set. Find $i_0 \in I$, and $i_1 \in I \setminus (i_0)$, and recursively find $i_n \in I \setminus (i_0, i_1, \dots, i_{n-1})$, this produces an ascending chain:

$$(i_0) \subsetneq (i_0, i_1) \subsetneq (i_0, i_1, i_2) \subsetneq \dots$$

Conversely, suppose every ideal in R is finitely generated, then for any ascending chain of ideals:

$$I_0 \subset I_1 \subset I_2 \subset \dots$$

their union I is also an ideal, which is finitely generated with its elements living in some I_N , thus the chain stabilises at this ideal. $\square$

Proposition 1.2.3. A principal ideal domain is a unique factorisation domain.

Proof. First we show that any $r \in R$ a PID can't be an infinite product of irreducibles. Suppose such an r exists, then as r irreducible we can find a nontrivial factorisation $r = r_1 s$, one of these must contain infinitely many irreducibles, let this be r_1 . Repeat this process ad infinitum to get a chain of ideals

$$(r) \subsetneq (r_1) \subsetneq (r_2) \subsetneq \dots$$

as R is a PID it is also Noetherian, so this ascending never stabilising chain of ideals cannot exist.¹

We now wish to show any such product of irreducibles

$$r_1 r_2 \dots r_n = r = s_1 s_2 \dots s_m$$

are equivalent. First note that in a PID an element is irreducible iff it is prime.² We can now see each r_i appears in the RHS, as we are living in an integral domain we can cancel and thus are done. $\square$

1.3 Unique factorisation in polynomial rings

Definition 1.3.1. Let R be a UFD, if $f \in R[x]$ the **content** of f is the gcd of its nonzero coefficients, written $c(f)$.

Lemma 1.3.2. If $f, g \in R[x]$, then $c(fg) = c(f)c(g)$.³

This really follows easily from

Lemma 1.3.3. If $c(f) = c(g) = 1$ then $c(fg) = 1$.

Proof. Suppose $c(fg) \neq 1$, then there exists some prime p dividing all the coefficients. First observe that as R is an integral domain, so is $R[x]$. Now consider the projection mod p , as $c(f) = c(g) = 1$ there exists no prime dividing all elements so they are both nonzero in $R/(p)[X]$ which we know to be an ID, but supposedly their product is 0. This is clearly a contradiction. $\square$

For our next step we need the following

Theorem 1.3.4 (Gauss lemma). If R is a UFD, and $f \in R[x]$ splits completely in $\text{Frac}(R)[x]$, then f splits completely in $R[x]$.

From these two statements you should apparently be able to deduce that if R is a UFD, so is $R[X]$???

¹This proof is clearly nonconstructive as we are choosing countably many irreducible factorisations all at once

²Given r irreducible, if $r \mid ab$ we can consider $c = \gcd(r, a)$ and observe $c \mid r$. I'm only really interested in R an ED, so lets assume that. We know $r = cd$ for some d , as r is irreducible either c is a unit, in which case we can write $xr + ya = c$ implying $xrb + yab = cb$, as $r = ab$ we have that $r \mid b$. If, instead, c is an associate of r , then $r \mid a$. The other direction is easier.

³As with lots of things in this course, this must be taken up to multiplication by a unit.

1.4 Newton polytopes

1.5 Hilbert basis theorem

2 Invariant theory

All rings are commutative.

Definition 2.0.1. If K is a field, a k -**algebra** is a ring R containing k . If $k \subset R_1, R_2$ are two such k -algebras, then a k -**algebra homomorphism** is a ring homomorphism which is the identity when restricted to k .

$R = \mathbb{C}[X_1, \dots, X_n]$ is a $\mathbb{C}$ algebra, $\mathbb{Z}$ is not a k -algebra for any k .

Definition 2.0.2. A k -algebra R is **finitely generated** if there exists a finite subset $S\{a_1, \dots, a_n\} \subset R$ such R is the smallest k -subalgebra containing S .

Equivalently, if there exists a k -algebra homomorphism

$$\text{ev}_S : k[X_1, \dots, X_n] \rightarrow R$$

which send f to $f(a_1, \dots, a_n)$ and is surjective.

So all f.g. k -algebras are rings of the form

$$k[X_1, \dots, X_n]/I$$

for some ideal I which, by the Hilbert basis theorem, will be finitely generated. We like these, because *in principle* every questions about f.g. k -algebra (of which there are many of great interest) can be answered algorithmically.

Even though there are many interesting k -algebras which aren't finitely generated, like $\mathbb{C}[[X]]$ the ring of power series in complex coefficients with radius of convergence $R > 0$, so we can do pointwise operations.

The group $GL_n(\mathbb{C})$ has a **right**⁴ action on $\mathbb{C}[X_1, \dots, X_n]$ by acting on $\mathbb{C}^n$ as matrices before applying the polynomial.

Definition 2.0.3. A **graded k -algebra** is a k -algebra

$$R = \bigoplus_{n \in \mathbb{N}} R_n$$

such that $R_0 = k$ and for each $n \geq 1$, R_n is a finite dimension k -vector space, and for all $n, m \in \mathbb{N}$, $R_n R_m \subset R_{n+m}$.

A **graded k -algebra homomorphism** $f : R \rightarrow S$ should satisfy $f(R_n) \subseteq S_n$ for all $n \in \mathbb{N}$.

$k[X_1, \dots, X_n]$ is the obvious example of a graded k -algebra by the degree of the polynomials, and the action of $GL_n(k)$ is in fact graded, as it preserves polynomial degree.

2.1 Finite generation

Proposition 2.1.1. For $G \leq GL_n(\mathbb{C})$ a finite subgroup acting on $R = \mathbb{C}[x_1, \dots, x_n]$, the set of G -invariant polynomials R^G is a ring.

Proof. Follow your nose. □

Theorem 2.1.2. Under the same assumptions, R^G is in fact finitely-generated as a $\mathbb{C}$ -algebra.

First proof. Let $I \trianglelefteq R$ be the ideal generated by G -invariant polynomials of positive degree. By Hilbert's basis theorem, I is finitely generated by some $r_1, \dots, r_m \in R$. Each r_i is of the form

$$r_i = \sum f_{ij} a_{ij}$$

with each $f_{ij} \in R$ and $a_{ij} \in R^G$. R^G is now generated by the a_{ij} . □

⁴People really don't like right actions, they like turning them into left actions by consider G^{op} or explicitly acting by inverses, we want to be able to efficiently compute things involving our actions, so we will tolerate just working with right actions.

We can actually improve on this. As the action of G on R is *graded*, we take our finite generating set to contain only homogeneous polynomials.

This proof is very concise, and very clean. But it is basically useless, as it is not *effective*! We have no way to actually produce the finitely many generators of R^G , just the incorporeal promise of their existence somewhere. We should find a **better** proof.

Definition 2.1.3. We have the **Reynold's operator**:

$$\rho : R \rightarrow R^G \quad \rho(f) = \frac{1}{|G|} \sum_{g \in G} f^g$$

basically an average, and obviously a ring homomorphism.

Lemma 2.1.4. The Reynold's operator satisfies the following properties:

1. $\rho(G) \subseteq R^G$;
2. $\rho|_{R^G} = \text{id}$;
3. ρ is R^G -linear;
4. ρ preserves the grading of R

Proof. There are all quite direct. □

Theorem 2.1.5. R^G is generated by the set of all elementary symmetric polynomials on $\{x_i^g\}_{g \in G}$ for each variable i , and $\rho(x_1^{\alpha_1} x_2^{\alpha_2} \cdots x_n^{\alpha_n})$ for all $0 \leq \alpha_1, \dots, \alpha_n \leq |G|$.

Proof. First we show that R is generated by all the elementary symmetric polynomials on $\{x_i^g\}_{g \in G}$ and the same monomials $x_1^{\alpha_1} x_2^{\alpha_2} \cdots x_n^{\alpha_n}$ with highest power $\leq |G|$, without applying the Reynold's operator.

For each x_i , consider the polynomial:

$$P_i(T) = \prod_{g \in G} (T - x_i^g) = T^{|G|} + a_{i,1} T^{|G|-1} + \cdots + a_{i,|G|}$$

the coefficients $a_{i,j}$ are exactly all the elementary symmetric polynomials on $\{x_i^g\}_{g \in G}$. For each i , $P_i(x_i) = 0$, thus:

$$x_i^{|G|} = -a_{i,1} x_i^{|G|-1} - \cdots - a_{i,|G|}$$

Hence, by induction, we can always strictly decrease the degree of any polynomial $f \in R$ until it is just the sum of monomials $x_1^{\alpha_1} x_2^{\alpha_2} \cdots x_n^{\alpha_n}$ with highest power $\leq |G|$.

Now, for any $f \in R^G$, we write:

$$f = \rho(f) = \rho\left(\sum a_{ij} x_1^{\alpha_1} \cdots x_n^{\alpha_n}\right) = \sum a_{ij} \rho(x_1^{\alpha_1} \cdots x_n^{\alpha_n}) \quad \square$$

2.2 Symmetric polynomials

3 Modules

3.1 Free modules

3.2 Modules over a PID

4 Matrix Lie groups

Formally, Lie groups are the group objects in the category of smooth manifolds. Many interesting examples and much deep theory of Lie groups is accessible by only considering matrix Lie groups, which avoid developing the full theory of smooth manifolds.

Definition 4.0.1. For a commutative ring R , we write:

$$GL_n(R) := \{n \times n \text{ invertible matrices with coefficients in } R\}$$

and we will almost always only be interested in $GL_n(\mathbb{R})$ and $GL_n(\mathbb{C})$.

These are both topological spaces by considering the inclusion:

$$GL_n(\mathbb{R}) \subset M_{n \times n}(\mathbb{R}) = \mathbb{R}^{n^2}$$

$GL_n(\mathbb{R})$ is in fact an open subset, as the determinant:

$$\det : M_{n \times n}(\mathbb{R}) \rightarrow \mathbb{R}$$

is a polynomial in its coefficients, so must be continuous. We can thus write

$$GL_n(\mathbb{R}) = \det^{-1}(\mathbb{R}^\times)$$

so $GL_n(\mathbb{R})$ inherits a locally Euclidean topology from $\mathbb{R}^{n^2}$.⁵

Definition 4.0.2. A **matrix Lie group** is a closed subgroup of $GL_n(\mathbb{R})$.

Some classical examples of matrix Lie groups include:

$$(\mathbb{R}, +) \cong \left\{ \begin{pmatrix} 1 & \lambda \\ 0 & 1 \end{pmatrix} \in GL_n(\mathbb{R}) \right\} = a^{-1}(1) \cap c^{-1}(0) \cap d^{-1}(1)$$

$$O_n(\mathbb{R}) = \{A \mid A^T A = I\} \subset GL_n(\mathbb{R}) \quad (\det(O_n) = \pm 1)$$

$$SL_n(\mathbb{R}) = \det^{-1}(1) \subset GL_n(\mathbb{R})$$

$$U_n(\mathbb{C}) = \{A \in GL_n(\mathbb{C}) \mid A^\dagger A = I\} \quad (\det(U_n) = S_1)$$

$$\text{Eucl}_n^+(\mathbb{R}) = \mathbb{R}^n \rtimes SO_n(\mathbb{R})$$

$$\text{Aff}_n(\mathbb{R}) = \mathbb{R}^n \rtimes GL_n(\mathbb{R})$$

$$\text{finite reflection groups like } D_{2n} \leq O_{2n}$$

$$\text{any finite group } G \leq S_n \leq SL_n$$

$$O(p, q) = \{A \in GL_{p+q}(\mathbb{R}) \mid \text{such that } A \text{ fixes all quadratic forms with inertia } (p, q)\}$$

4.1 Matrix exponential

Definition 4.1.1. If $A \in M_{n \times n}(\mathbb{R})$ define the matrix exponential:

$$\exp(A) = \sum_{k=0}^{\infty} \frac{1}{k!} A^k$$

Proposition 4.1.2. For all matrices $A \in M_{n \times n}(\mathbb{R})$ this converges

Sketch. As all norms on $\mathbb{R}^{n^2}$ are equivalent, we can show convergence by any of them, we'll use the d_∞ norm. Going by absolute convergence

$$\left\| \sum_{k=0}^n \frac{1}{k!} A^k \right\| \leq \sum_{k=0}^n \frac{1}{k!} \|A^k\| \rightarrow e^{\|A\|}$$

gives us our bound, we then show the sequence is Cauchy. □

Theorem 4.1.3. We can in fact do even better. Not only is $\exp$ continuous, it is continuously differentiable, and all derivatives converge uniformly on all compact subsets of the domain.

Some obvious first properties of the matrix exponential

⁵The same argument applies for $GL_n(\mathbb{C})$, I often won't include this in the future.

4.2 Lie algebras

Theorem 4.2.1. $f : G_1 \rightarrow G_2$ a continuous group homomorphism of matrix Lie groups induces a homomorphism of Lie algebras $f_* : \mathfrak{g}_1 \rightarrow \mathfrak{g}_2$.